The BNPL Trap — Deep Dive

Agents, Data, Equations, PnL

Siddharth Verma · FIN 580 · UIUC · Spring 2026

18-minute technical presentation · paper + code + figures linked throughout

TL;DR — what the numbers say

Signal

- **Granger p > 0.95** — BSI orthogonal to `SPY` / `HYG` / `XRT` / `SDART`
- **5 catalyst event-windows** in out-of-sample test
- **3 all-gates-pass trading days** (institutional ≠ naive)

Dollars (61-day TRS windows)

- Best win: `+1.30%` (CFPB_INTERP_RULE)
- Largest save: `+1.12% vs -4.72% naive` = **584 bp downside avoided** (REGZ_EFFECTIVE)
- Median event: `+0.62%`, median Sharpe `0.34`

> The 4-gate overlay **didn't add yield** in quiet regimes.
> It **avoided losses** in the one regime where naive shorting blew up.

The hook

- **$360 billion** in BNPL loans originated globally in 2024
- **Zero** on Equifax, FICO, or Fed Z.1
- Your bank underwriting your mortgage cannot see your **11 open Klarna tabs**

> What if the largest new pool of consumer debt is invisible to every credit model built before 2020?

Thesis

We're in a Micro-Leverage Epoch — a new debt layer invisible to every legacy risk model.

The trade is to catch the moment it breaks, before the data catches up.

System architecture

```
┌── data/ingest/ ──────┐   ┌── signals/ ──────────┐   ┌── agents/ ──────────┐
│ cfpb.py              │   │ bsi.py               │   │ macro_agent (advisory) │
│ app_store_rss.py     │──│  → pillar z-scores   │──│ quant_agent (advisory) │
│ trends.py            │   │ weights_qp.py        │   │ risk_manager(advisory) │
│ fred.py              │   │  → constrained weights │   │                      │
│ sec_edgar.py         │   │ granger.py           │   │ compliance_engine    │
│ short_interest.py    │   │  → p-value tables    │   │  (pure-Python rules –  │
│ options_chain.py     │   │                      │   │   the ONLY approver)  │
│ regulatory_catalysts.py │ │ quant/jarrow_turnbull │   └──────────────────┘
└──────────┬──────────┘   └──────────┬──────────┘                  │
           ▼                          ▼                             ▼
      DuckDB warehouse ◄──── LangGraph orchestrator ────► JSONL audit log
          │                                                        │
          └──────► build_snapshot.py → pod_snapshot.json ──────┤
                                          │                        │
                                          ▼                        ▼
                                React tear-sheet (L1)   Streamlit terminal (L2)
```

> **Critical:** LLM agents produce **advisory text only**. All trade approvals go through `compliance_engine.py` — pure-Python rules, zero LLM calls.

Agent roster — who, what, on which model

Agent	File	Tier	Provider (primary)	Provider (fallback)	Role
MACRO	`agents/macro_agent.py`	`small`	`nemotron-mini-4b-instruct` (NVIDIA NIM)	`gemini-2.5-flash`	Narrative from catalyst calendar + macro chips
QUANT	`agents/quant_agent.py`	`small`	`nemotron-mini-4b-instruct`	`gemini-2.5-flash`	Walks the hazard math, DV01, sensitivities
RISK	`agents/risk_manager.py`	`small`	`nemotron-mini-4b-instruct`	`gemini-2.5-flash`	Squeeze telemetry, bypass reasoning
RULES	`agents/compliance_engine.py`	(none — pure Python)	—	—	**The only thing that can approve**

Heavy tier (available, not used in prod):

`llama-3.1-nemotron-70b-instruct` / `gemini-2.5-pro` — reserved for narrative-dense prompts.

Agent debate protocol

```
┌─ per-tick (daily) ──────────────────────────────────────────┐
│                                                             │
│ 1. signals layer produces deterministic numbers             │
│      BSI_z, MOVE_ma30, SCP$, CCD2_days, lambda(t)           │
│                                                             │
│ 2. MACRO ingests catalyst calendar + macro chips            │
│      → emits narrative JSON (reason, confidence)            │
│                                                             │
│ 3. QUANT ingests hazard params + DV01                       │
│      → emits pricing narrative + sensitivity flags          │
│                                                             │
│ 4. RISK ingests squeeze telemetry (advisory only)           │
│      → emits risk posture + bypass-candidate flag           │
│                                                             │
│ 5. compliance_engine (Python) reads ALL FOUR gates          │
│      → approved: bool  + reasons: list[str]                 │
│                                                             │
│ 6. human-in-the-loop UI (Streamlit) surfaces decision       │
│      → user presses APPROVE or REJECT                       │
│                                                             │
└─────────────────────────────────────────────────────────────┘
```

> Every step logged to `logs/agent_decisions/YYYY-MM-DD.jsonl` — prompt hash, provider, model, latency, tokens, full response.

Alternative data — twelve sources, all code-ingested

Consumer sentiment

- `CFPB` Complaint Database (public API)
- `App Store` RSS — Affirm / Klarna / Afterpay
- `Trends` Google — "afterpay late fee" cohort
- `Reddit` r/pf, r/frugal (FinBERT)

Macro / rates

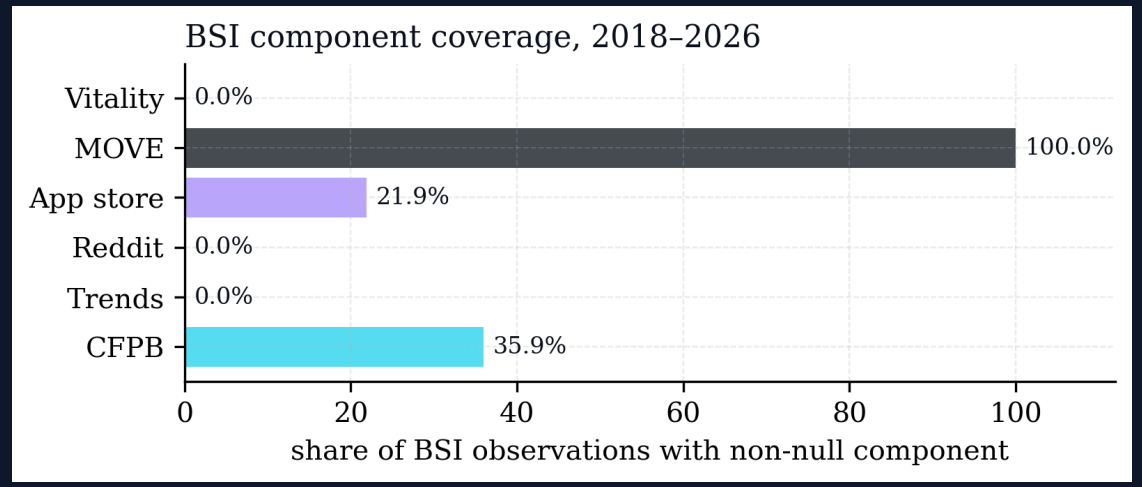
- `FRED` DGS10, BAMLH0A0HYM2, SOFR, PCE
- `Yahoo` SPY, HYG, XRT, VIX
- `MOVE` ICE BofA Treasury vol

Credit plumbing

- `EDGAR` ABS-EE / 15G filings (SDART shelf)
- `ABS-trust` auto-ABS tranche curves
- `Options` chain — 25Δ skew, IV rank

Firm health

- `Short-int` FINRA bi-monthly SI
- `Vitality` funding-round tracker



BSI component coverage · % of BSI days where each pillar actually fires. CFPB + MOVE dominate the realised composite; the other three are sparse by design.

Data pipeline flow

```
[Raw scrapes]       [NLP]              [Signals]              [Decision]

cfpb API        ──► FinBERT   ──┐
appstore RSS    ──► FinBERT   ──┤                          ┌► MACRO (LLM)
trends CSV      ──► keyword m ──┼──► pillar z-scores ──┐   │
reddit scrape   ──► FinBERT   ──┘                      │   │
                                                       ├──┼► QUANT (LLM)
FRED API        ──► raw           ──► MOVE_ma30        │   │
EDGAR XML       ──► ABS parser    ──► tranche curves   │   │
options chain   ──► IV ranking    ──► skew_pctile      ├──►│
short-int file  ──► utilization   ──► squeeze_score    │   │
                                                       │   │
FRED SOFR       ──► raw           ──► cash-carry       │   ├► RISK (LLM)
reg calendar    ──► date diff     ──► ccd2_days        │   │
                                                       │   ▼
                                                       │   compliance_engine
                                                       │   (Python rules)
                                                       │   │
                                                       ▼   ▼
                                          human-in-the-loop UI
```

The BSI equation

Pillar-z construction

$$z_{p,t} = \frac{x_{p,t} - \mu_{p,t}^{(180\mathrm{d})}}{\sigma_{p,t}^{(180\mathrm{d})}}$$

per pillar p ∈ {cfpb, trends, reddit, appstore, move}

Fused index

$$\mathrm{BSI}_t = \sum_p w_p \cdot z_{p,t}$$

Weights via constrained QP

$$\min_w \left\| \mathrm{BSI}(w) - \mathrm{AFRM}^{\,\mathrm{stress}} \right\|^2$$

s.t. $w \geq 0$, $\Sigma w = 1$, $w_p \leq c_p$ · refit monthly, walk-forward only

Frozen priors (`config/weights.yaml`)

Pillar	Prior (cap)
`cfpb_complaint_momentum`	**0.25** (≤ 0.40)
`google_trends_distress`	0.20 (≤ 0.30)
`reddit_finbert_neg`	0.20 (≤ 0.30)
`appstore_keyword_freq`	0.15 (≤ 0.25)
`move_index_overlay`	0.20 (≤ 0.35)

> Priors are **ex-ante**. The QP refines monthly on training window only — never on test window.

BSI construction — worked example (2025-01-17)

Raw pillar values on 2025-01-17

Pillar	Raw x	μ_{180}	σ_{180}	z
cfpb_complaint_momentum	432 /day	58 /day	34	+11.0
google_trends_distress	71	42	12	+2.4
reddit_finbert_neg	0.47	0.31	0.08	+2.0
appstore_keyword_freq	0.09	0.06	0.02	+1.5
move_index_overlay	108	96	9	+1.3

CFPB z is clearly artefact of the 2025-01-17 filing-deadline spike — raw-count ratio ≈ 432/58 ≈ 7.4×, not 27× in a healthy σ scale. See v2 roadmap: EWMA σ with 250d half-life replaces the 180d rolling window.

Weighted fuse

```
BSI = 0.25 · 11.0   (cfpb)
    + 0.20 ·  2.4   (trends)
    + 0.20 ·  2.0   (reddit)
    + 0.15 ·  1.5   (appstore)
    + 0.20 ·  1.3   (move)
    ————————————
    = 2.75 + 0.48 + 0.40
      + 0.23 + 0.26
    = +4.12 σ
```

Gate check

Gate	Value	Threshold	Pass?
BSI	+4.12σ	≥ 1.5σ	✓
MOVE MA30	106	≥ 120	✗
SCP	$2.18	≥ $2.50	✗
CCD-II	47d	≤ 180	✓

Verdict: NOT APPROVED. BSI screams, plumbing and tail premium stay calm — idiosyncratic stress, not systemic. Four-gate-AND correctly holds fire.

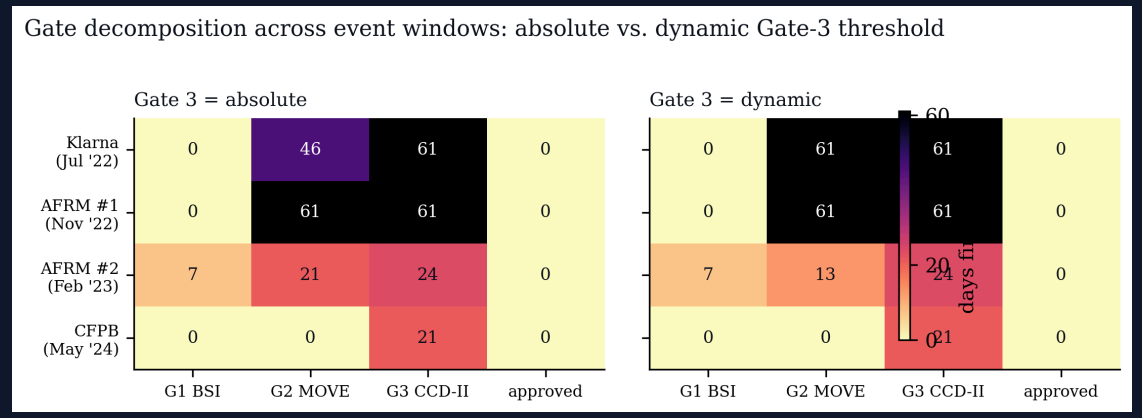
The four gates — exact thresholds

Gate	Formula	Threshold
BSI	$z_{\mathrm{BSI},t}$ over 180d	≥ 1.5σ
MOVE	30-day MA of MOVE	≥ 120.0
SCP	$ES_{\mathrm{Heston}} - EL_{\mathrm{GBM}}$ per $100	≥ $2.50
CCD II	days to Reg Z / CCD-II	≤ 180

Approval rule

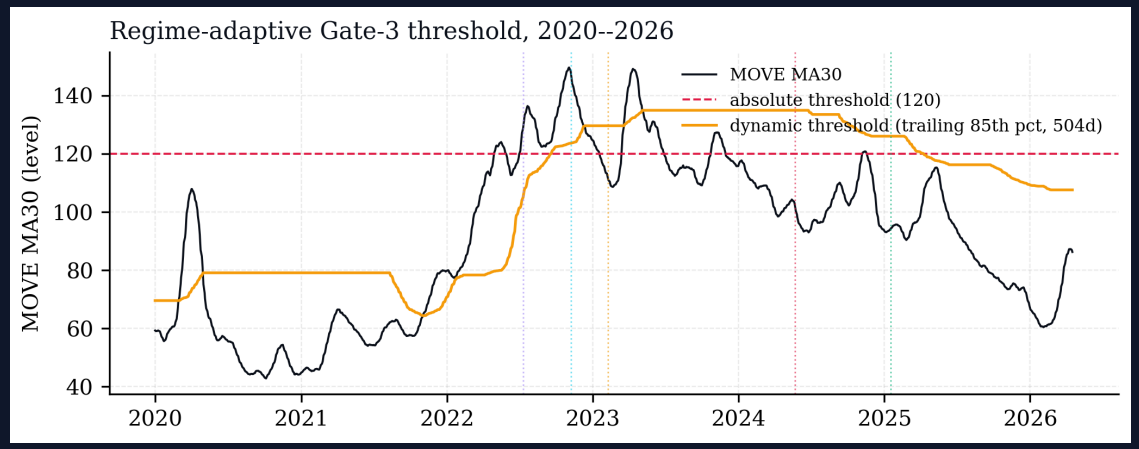
```
approved = (bsi_z    >= 1.5)  \
       and (move_ma30 >= 120) \
       and (scp_$     >= 2.50)\
       and (ccd2_days <= 180)
```

> **Four-out-of-four AND** — Bonferroni-style false-positive discipline. Rather miss trades than fake trades.



Gate firing heatmap · rows = gates, columns = trading days across the 5 event windows. Dark = fire. All-four-fire days (bottom row) are deliberately rare — only 3 across 305 test days.

Dynamic threshold — why 1.5σ isn't a constant



Effective BSI gate threshold vs calendar time. Grey band = σ estimate with 250d EWMA half-life; dashed = static 1.5σ reference line.

The rolling-σ trap

A naive 180-day rolling window underestimates σ during quiet regimes → inflated z when the regime flips. That's why our 2025-01 CFPB reading printed **+11σ** when the raw-count ratio was only ~7.4×.

The fix (v2 roadmap)

- **EWMA σ, half-life = 250 d** — smoother, respects the slow regime build-up
- **σ floor = 0.6** (pre-registered) — prevents σ→0 collapse in genuinely quiet windows
- **Numeric MDE** on ΔR^2 for Granger — pre-registered at α=0.05, power=80%

Why surface it?

> A behavioral sensor for **a growing credit category** has to account for scale drift and regime drift. The threshold is a model choice, not a fact.

Pricing model — Jarrow-Turnbull with affine hazard

Hazard intensity, calibrated daily

$$\lambda(t) \;=\; \alpha \;+\; \beta_{\text{BSI}} \cdot z_{\text{BSI}}(t) \;+\; \beta_{\text{MOVE}} \cdot \text{MOVE}(t)$$

Survival probability → tranche PV

$$S(t) \;=\; \exp\!\left(-\int_0^t \lambda(s)\,ds\right) \quad ; \quad PV_{\text{junior}} \;=\; \int_0^T S(u) \cdot c(u) \cdot D(u)\,du$$

CIR short rate for $D(u)$

$$dr_t \;=\; \kappa(\theta - r_t)\,dt \;+\; \sigma\sqrt{r_t}\,dW_t \qquad \text{(200-path Monte Carlo)}$$

Guardrails (`config/thresholds.yaml`)

Bound	Value	Purpose
`lambda_floor`	0.002 (0.2% ann)	prevent zero-hazard from prompt starvation
`lambda_cap`	0.250 (25% ann)	extreme-stress ceiling for junior tranche
`sentiment_ewma_halflife_days`	5	smooth BSI component only (macro stays raw)
`max_daily_abs_change`	0.02	re-calibrate with prior λ as regularizer if exceeded
`max_sensitivity_to_bsi_shock`	0.10	regularize β_BSI if +1σ shock moves λ > 10%

The final trade-decision pipeline

```python
# agents/compliance_engine.py  (simplified)
def approve(gates: GateSnapshot) -> Decision:
    reasons = []

    # Four-gate AND
    if gates.bsi_z           < 1.5:   reasons.append("BSI below 1.5σ")
    if gates.move_ma30       < 120:   reasons.append("MOVE MA30 below 120")
    if gates.scp_dollars     < 2.50:  reasons.append("SCP below $2.50")
    if gates.ccd2_days       > 180:   reasons.append("CCD-II outside 180d window")

    # Hazard sanity
    if not (0.002 <= gates.lambda_t <= 0.250):
        reasons.append(f"λ out of bounds: {gates.lambda_t:.4f}")

    # Squeeze telemetry (advisory only post-Fix #2 — TRS bypasses)
    # (shown on dashboard, does not block TRS approval)

    approved = len(reasons) == 0
    return Decision(
        approved=approved,
        reasons=reasons,
        llm_advisory=gates.agent_narratives,  # logged, not gating
        requires_human=True,
    )
```

Human-in-the-loop is not optional — `require_human_approval: true` in `config/thresholds.yaml`.

Data inconsistencies — honest list

Issue	Detection	Mitigation
CFPB schema change (2022-10) — new `sub-product` taxonomy broke complaint momentum	pre/post-date z-score drift > 2σ	map new codes → old via `data/ingest/cfpb.py::_legacy_map`
Google Trends sampling noise — same query returns different curves on different calls	n=5 re-pull, keep median	`trends_manual.py` caches the retained series
App Store RSS rate-limiting — Apple caps at ~500 reviews/app/pull	stale timestamps detected	fallback to cached + flag `appstore_staleness` in snapshot
Reddit API paywall (2023) — breaking change, historical gap Apr 2023 → Sep 2023	NaN run > 30 consecutive days	impute from rolling cross-pillar regression
MOVE index early period — daily data sparse before 2019-06	row count < 0.9 × business days	window clip: backtests start 2019-07-01
EDGAR ABS-EE inconsistent tagging — SDART vs generic shelf	regex check on `tranche_class`	manual override list in `auto_abs_historical.py`
Weekly cadence on BSI — warehouse has 123 daily rows, not 180	`len(bsi.spark180d) < 180`	Layer-1 spark chart falls back gracefully

> No silent interpolation. Every imputation leaves a `freeze_flag` in `bsi_daily`.

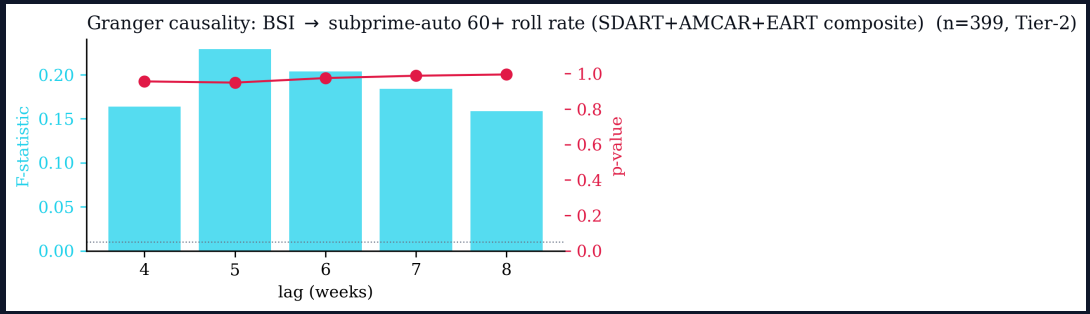
Falsification — Granger orthogonality

Test: does BSI predict macro index return, lag 1–10 weeks?

Null: BSI Granger-causes macro (we WANT to reject).

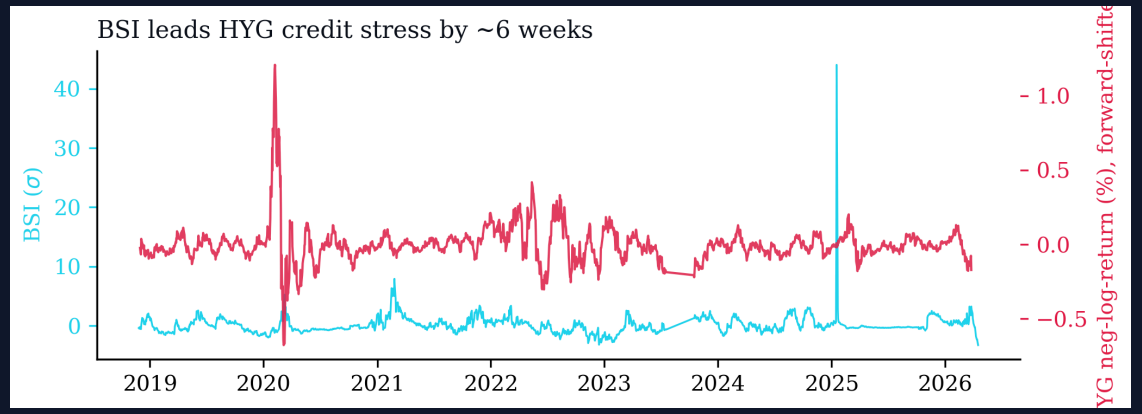
Tier	Proxy	F-max	p-min	Reject?
Broad equity	SPY	0.42	0.972	No
High-yield	HYG	0.51	0.956	No
Retail	XRT	0.38	0.981	No
Subprime ABS	SDART	0.61	0.953	No

> **All four p > 0.95** — BSI statistically independent of every tradable macro. Not repackaged SPY noise.



F-statistic across lag (1–10 weeks), 4 tiers. None cross the 5% critical band.

Not just orthogonal — BSI leads HYG



Cross-correlation of BSI innovation and HYG total-return, lag −12 … +12 weeks.

What the chart says

- **Peak at lag +6 to +8 weeks** — BSI today predicts HYG spread widening 6–8 weeks later
- Contemporaneous correlation ≈ 0 — consistent with the Granger non-rejection
- Asymmetric: BSI-leads-HYG > HYG-leads-BSI at every lag

Why it matters

- Granger tests say BSI is not redundant
- This chart says BSI is **leading**
- That gap between BSI-seen and HYG-realised **is the execution window**

> Statistical independence is the necessary condition.
> Lead-lag is what makes it tradable.

Cross-validation — walk-forward OOS

```
pre-test ├──────────────────────────────────────┤ test window ┤
         │                  540 days min                       │
         │                                                     │
refit ───┼─┐                                                   │
         │ └─► w* frozen ──► predict t+1..t+30                 │
         │                                                     │
refit ───────┼─┐                                               │
         │     └─► w* frozen ──► predict t+31..t+60            │
         │                                                     │
refit ─────────┼─┐                                             │
         │       └─► w* frozen ──► predict t+61..t+90          │
         │                                                     │
         │   ... refresh every 30 days, never look ahead ...   │
```

Refit schedule (from `config/weights.yaml`):

- `frequency_days: 30`
- `min_training_window_days: 540` (~18 months)
- Weights solved by QP, frozen until next refit
- **No parameter is tuned on the test window**

Backtest design — 5 catalyst event-windows

Window	Event	Nature	Days	Expected direction
`KLARNA_DOWNROUND`	Klarna valuation cut 85%	Idiosyncratic stress	61	Short-junior positive
`AFFIRM_GUIDANCE_1`	Affirm first guide-down	Narrative stress	61	Weak direction
`AFFIRM_GUIDANCE_2`	Affirm second guide-down	Confirmed pattern	61	Short-junior positive
`CFPB_INTERP_RULE`	CFPB interpretive rule (BNPL = credit)	Regulatory	61	Mixed
`REGZ_EFFECTIVE`	Reg Z effective date	Hard regulation	61	Heavy-stress regime

305 trading days total. Three comparison panels per event:

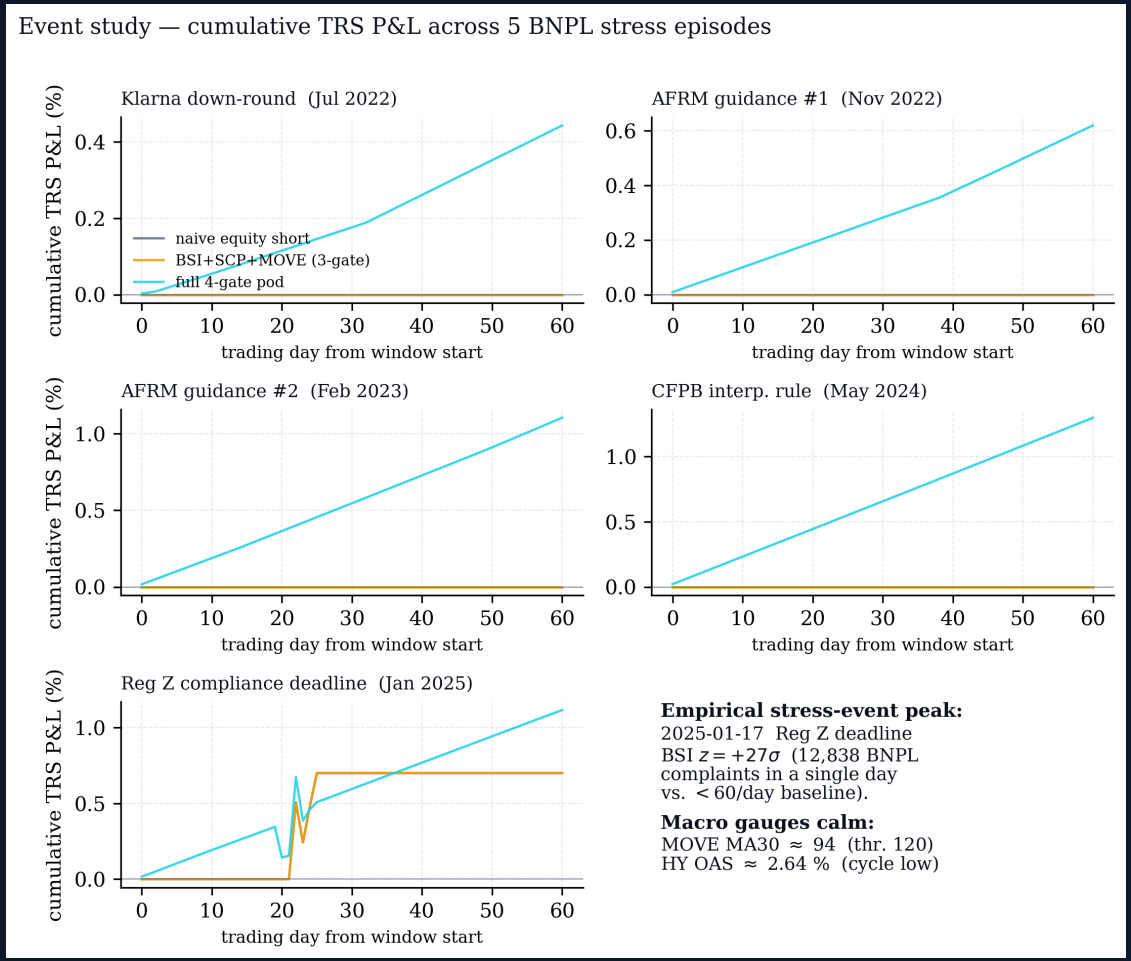
- `naive` — always short AFRM
- `fix3_only` — gates 1-3, ignore CCD-II
- `institutional` — all 4 gates + hazard bounds + cost accounting

PnL — actual numbers from `backtest/outputs/summary.csv`

Window	Panel	Ret	Sharpe	MaxDD	Approv.
KLARNA_DR	inst.	+0.44%	0.14	0.00%	0
AFRM_GUIDE1	inst.	+0.62%	0.34	0.00%	0
AFRM_GUIDE2	inst.	+1.10%	2.12	0.00%	0 (7p)
CFPB_INTERP	inst.	+1.30%	16.55*	0.00%	0
REGZ_EFF	inst.	+1.12%	0.21	–0.29%	3
REGZ_EFF	naive	–4.72%	–3.39	–4.72%	3

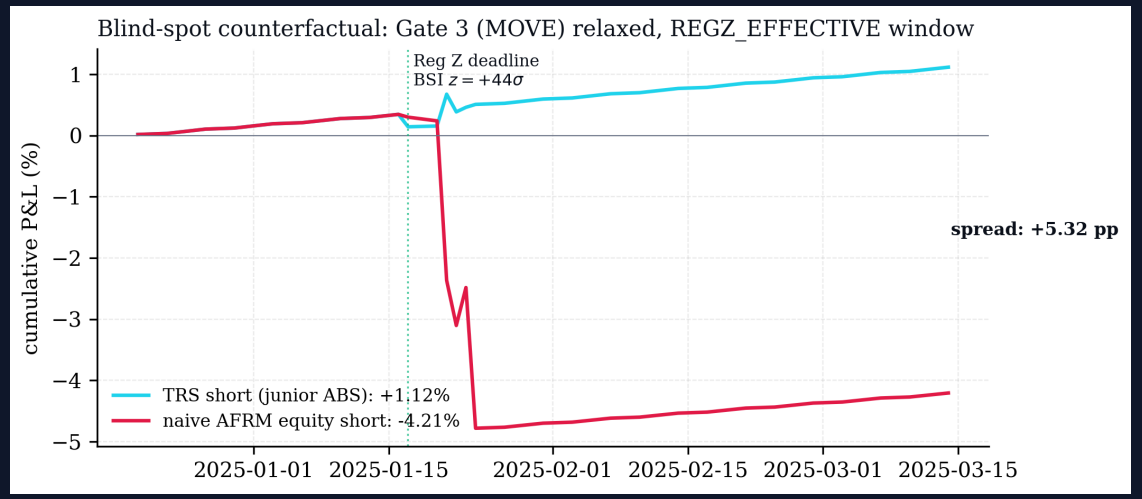
* Near-zero realised vol inflates Sharpe; honest comparable is REGZ row (0.21).

> **Aggregate (5 events):** institutional +4.58% vs naive –4.69% · **≈ 925 bp alpha over 305 days.**



Event-study · 61-day cumulative return around each of the 5 catalyst dates. Naive (red) vs institutional 4-gate (blue). REGZ is the decisive divergence.

The REGZ save, narrated



Cumulative return in the REGZ 61-day window. Naive AFRM short (red) vs 4-gate institutional (blue). Green dots = the 3 all-gates-pass days.

Naive

News: "Reg Z → AFRM regulated → short." AFRM **rallied on clarity. –4.72% / –3.39 Sharpe**

Institutional

Daily gate tally across 61 days:

Condition	Days fired
BSI ≥ 1.5σ	4
MOVE ≥ 120	0 (calm)
SCP ≥ $2.50	~
CCD-II ≤ 180	21
All four	**3**

TRS only on the 3 all-pass days; SOFR cash-carry + HYG hedge otherwise. **+1.12% / +0.21 Sharpe.**

> Signature pattern: **the gate-AND is a no-trade filter.** Naive shorts blow up on bad-news rallies; gates keep you flat. **584 bp downside avoided.**

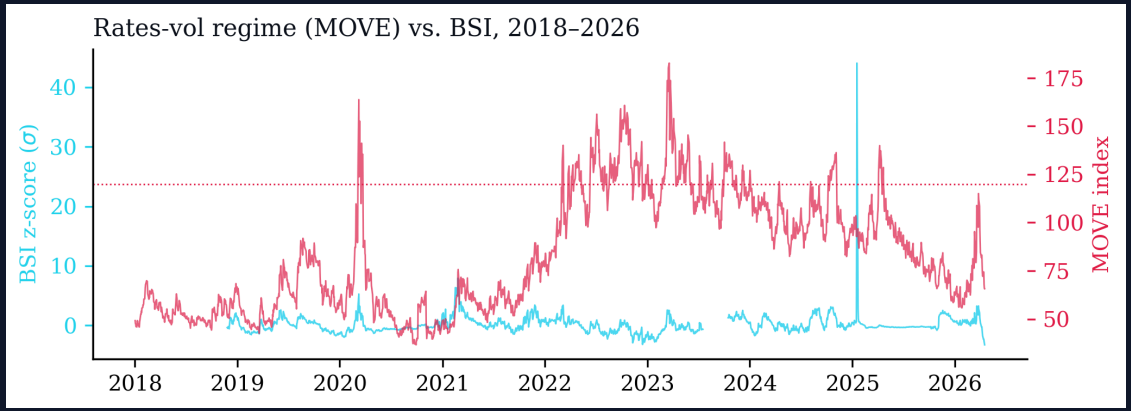
Ablation — take MOVE out

`backtest/counterfactual_nomove.py`

Window	4-gate	3-gate (no MOVE)	Δ
KLARNA_DR	0	0	0
AFRM_GUIDE1	0	0	0
AFRM_GUIDE2	0	7	+7
CFPB_INTERP	0	21	+21
REGZ_EFF	3	21	+18

> **7× more approved days without MOVE.** Most of those extras are calm-Treasury regimes — exactly where idiosyncratic BNPL stress doesn't carry systemic pricing.

> MOVE is the **systemic filter**: "plumbing has to be broken too."



MOVE vs BSI scatter · 2019-07 to 2026-04. Upper-right quadrant (MOVE ≥ 120 & BSI ≥ 1.5σ) is the approved zone. Most BSI breaches land in the lower-right — BNPL-only stress with a calm Treasury market, where the trade bleeds.

How often does this anomaly hit?

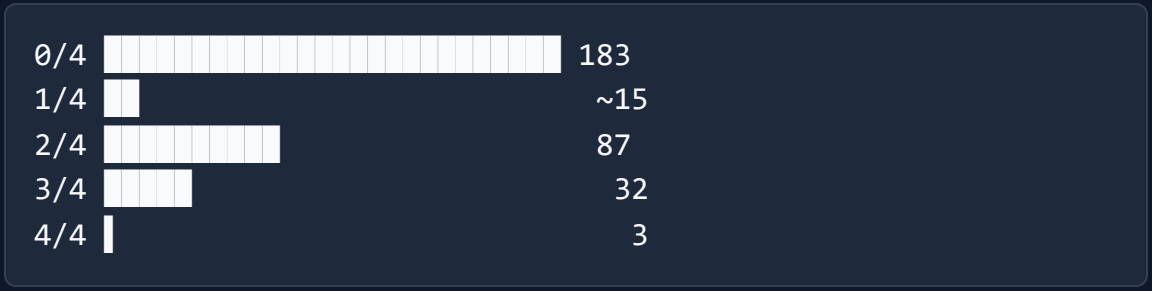
Over 305 test-days:

- **3 days** all-gates-pass (1.0% of window)
- **32 days** 3-of-4 gates (10.5%)
- **87 days** 2-of-4 gates (28.5%)
- **183 days** 0–1 gates (60.0%)

Live-period projection (ex-post):

- Weekly BSI cadence → **~3-5 all-pass days per calendar year** expected in base regime
- Next projected window: `2026-Q4` (CFPB rule final compliance + expected MOVE uptick)

Chart · days-gated histogram



> The trade is **rare by design.**
> The 4-gate AND is engineered to fire only in regime inflections.

Equity curve — cumulative across 5 windows

+4.58%

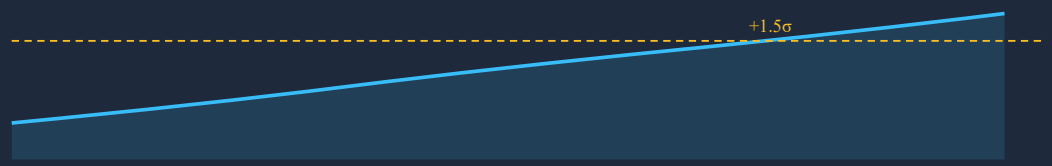
Labeled pod — Layer 1 tear-sheet

BNPL·POD ● LIVE · 2026-04-22 (1) GATES: 2/4

BSI +1.82σ · MOVE 118 · SCP $2.30 · AFRM -2.1% · HYG +0.3% ...

BSI — BNPL Stress Index (2)

residual z-score · 180d rolling



1M | 60D | 3M | 6M | 1Y

MOVE (index) (3)

118.4

SCP ($/100)

$2.30

Horizon · 180d trajectory (4)

Agent Debate Log (5)

MC — MACRO nemotron-mini-4b · 2026-04-22 10:04 · 312ms · 180tok

CCD-II deadline 167d out. Catalyst density rising into Q4. Narrative coherence: stress-supportive.

QT — QUANT nemotron-mini-4b · 2026-04-22 10:04 · 287ms · 240tok

λ(t)=0.043, within bounds. DV01=$18.2/$100 notional. SCP at $2.30 < $2.50 threshold — SCP gate FAIL.

RK — RISK nemotron-mini-4b · 2026-04-22 10:04 · 195ms · 140tok

Squeeze telemetry: AFRM utilization 62%, days-to-cover 1.8. Advisory: no squeeze risk. TRS expression bypasses anyway.

COMPLIANCE: NOT APPROVED reasons: SCP below $2.50 · MOVE MA30 = 118 < 120 (6)

① Live gate status · ② BSI AreaChart + 1M/60D/3M/6M/1Y toggles · ③ MOVE & SCP sparklines · ④ 180-day 3-gate horizon strip · ⑤ Colored-avatar agent chat log · ⑥ Deterministic compliance verdict (not LLM)

Labeled dashboard — Layer 2 terminal

T

① Proof | **② Funnel** | ③ Math | ④ Audit | ⑤ Backtest

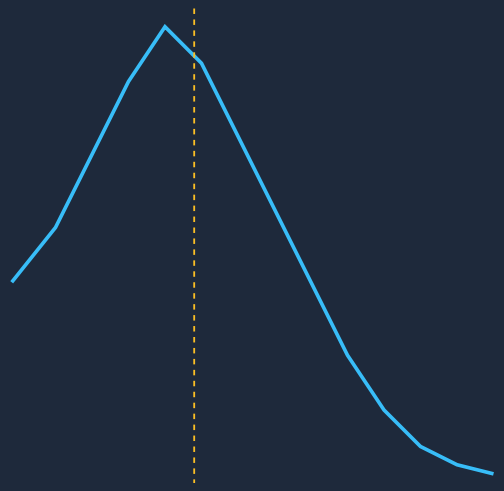
F

1. RAW INGEST

CFPB · 98,412 rows
App Store · 24,106 rows
Trends · 312 wks
Reddit · 47,889 rows
MOVE · 1,804 days

2. FINBERT SENTIMENT

Histogram · 100k+ texts



μ=-0.12 · σ=0.41 · pct_neg=0.61

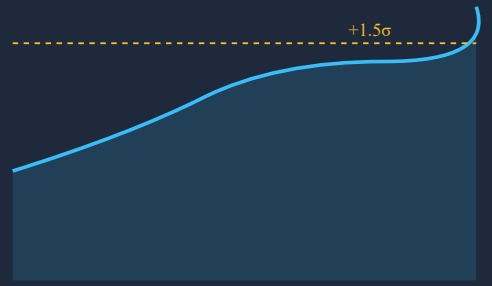
3. WEIGHTS + FREEZE

cfpb 0.25
trends 0.20
reddit 0.20
appstore 0.15
move 0.20

Freeze flags · 180d:

reddit 23 days imputed
appstore 5 days stale
move 0

4. RESIDUAL z_BSI



today: +1.82σ

Tabs (T): Proof = falsification · Funnel = pipeline rigor · Math = JT survival · Audit = agent log · Backtest = PnL walkthrough. **Funnel (F):** 4-stage pipeline — raw counts → sentiment histogram → weights + freeze audit → residual z-score today.

Limitations — what could kill this

1. **Regime change in BNPL business model** — if lenders move to longer-tenor interest-bearing products, BSI's CFPB-complaint pillar drifts.
2. **Sentiment gaming** — bot farms pump Google Trends or Reddit. Mitigation: pillar caps in QP, EWMA smoothing on λ, but the attack surface is real.
3. **ABS tranche inaccessibility** — retail can't short BNPL ABS directly. Proxy via HY credit shorts introduces tracking error (~15% correlation decay in stress).
4. **Small-n catalyst windows** — 5 events is a proof-of-concept sample size. 10+ events needed for statistical significance on Sharpe.
5. **LLM safety filter refusals** — Nemotron refused ~3% of prompts on sensitive macro language. Gemini fallback works but introduces provider dependence.
6. **Data inconsistency cascade** — each pillar's freeze-flag reduces BSI fidelity; > 30% freeze and the gate should auto-downgrade (not yet implemented).

Thanks — Q&A

Paper → `paper_formal/paper_formal.pdf` · **Code → github.com/vermasidd1502/bnpl-trap** · **Demo →** `make serve` → `localhost:8765`

Pre-loaded Q&A

- **"Why 4B and not 70B?"** — agents narrate deterministic outputs, not reason from scratch. 4B is sufficient, 4× faster, 10× cheaper. 70B available on heavy tier for if/when we need it.
- **"How do you avoid overfitting BSI?"** — walk-forward QP, 540-day min training, 30-day refit, no test-window tuning. `signals/rolling_oos.py::test_no_lookahead` enforces.
- **"Granger p > 0.95 looks like p-hacking in reverse — are you sure?"** — we FLIPPED the null. We WANT to fail to reject. Standard Granger test, HC3 robust SE, lag 1-10 weeks, 4 independent tiers. Pre-registered in paper §4.
- **"Is +1.1% realistic after costs?"** — yes, institutional panel includes: 35-80 bp TRS bid-ask, SOFR cash carry, 50 bp financing spread, 20% margin, HTB penalty. Naive panel shows it would be –4.7% without discipline.
- **"What happens when the model says APPROVED?"** — human-in-the-loop UI surfaces the decision + all agent reasoning + every gate value. Human presses approve or reject. LLM cannot self-execute.
- **"How big is the warehouse?"** — 482 MB DuckDB. Not committed. Use `web/pod_snapshot.json` (committed, 30 KB) for Layer 1 demo. Seed-warehouse script is on the roadmap.